

ARI2201 - Benchmarking Small-Object Detection on Brain MRI: A Cerebral Microbleed Case Study

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Abstract:

Cerebral microbleeds (CMBs) are critical biomarkers for various cerebrovascular diseases, yet detecting them in MRI scans is a very time-consuming and error-prone process in a field where precision is essential. This project aims to automate CMB detection using the nnU-Net deep learning framework on the VALDO Challenge 2021 dataset. Although standard evaluation metrics prove to be helpful with getting an overview of the model performance, they often fail to correctly capture spatial context, especially for small-object detections like CMBs, where a single misplaced voxel can heavily penalise the Dice score even though the model prediction would be correct. To address this issue, an Interactive Exploration Dashboard was developed using the Streamlit Python framework. The interface enables dynamic probability threshold adjustments and provides colour-coded overlays of True Positives, False Positives, and False Negatives directly on MRI slices for each subject. The results conclude that the model correctly detects the majority of microbleeds, while the dashboard proves to be an essential tool for improving result transparency and supporting the evaluation process.

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1 Introduction

Cerebral micro-bleeds (CMBs) are very small deposits of blood near the blood vessels of the brain. The presence of CMBs is most common with patients who suffer from cognitive diseases (such as stroke and dementia), as well as with individuals of higher ages. The accurate detection of the count and location of such phenomena is clinically significant as they could be conclusive markers to cerebrovascular diseases such as recurring strokes, cognitive decline and cerebral small vessel diseases [7]. Current detection methods involve radiologists manually sifting through hundreds of MRI slices to visually identify each CMB, a process which is prone to a high level of human error [15]. In fact, two radiologists doing CMB detection on the same subject will often disagree with the counts.[3]

1.1 Project Motivation and Data Preprocessing Pipeline

The main aim of this project is to automate the detection of CMBs using deep learning techniques, which would enable further research on their presence in the area of neurodegenerative diseases [3]. To establish a strong baseline, the methodology made use of the VALDO Challenge 2021 training dataset, which consists of 72 subjects with T1, T2, and T2*-weighted MRI volumes as well as their corresponding binary segmentation masks.[2]

Before starting the model training process, a data preprocessing pipeline was implemented, which was constructed of data curation, affine alignment and label verification. Data curation was done to adjust the structure of the entire dataset to fit the deep learning framework, as it required a strict folder structure and metadata. Volume alignment with the segmentation masks is critical to have reliable results from the trained model. Affine alignment was implemented using the $T2^*$ weighted image as the spatial reference to align the label masks. Furthermore, label verification was implemented to confirm that image dimensions match their corresponding labels, validate the integrity of binary masks, and calculate the initial connected components to assess the volume of individual microbleeds before training.

1.2 Model Training and Inference Pipeline

Following the preprocessing pipeline, the deep learning model was trained. The nnU-Net semantic segmentation framework was configured and trained using a 2D full-resolution

convolutional configuration with a 5-fold cross-validation across the VALDO dataset [6].

From the trained model, a robust inference and evaluation pipeline was developed. The pipeline performs inference on unseen $T2^*$ weighted volumes to generate the prediction files and probability maps (one per subject). Due to this data being meaningless on its own, the evaluation segment performs connected-component analysis to compare predicted lesions against ground-truth and output metrics such as lesion-level sensitivity, false positives per subject, F1-scores, and voxel-level Dice coefficients.

1.3 The Interactive Exploration Dashboard

Although basic metrics provide a helpful numerical summary of the model’s performance, they fail to capture how the model is actually working and the specific contexts of the model’s successes and failures. Due to this, my separate contribution to this project is the design and implementation of an Interactive Exploration Dashboard. Built using the Streamlit framework [14], this interface is designed to visualise the model’s inference performance and be as informative as well as easy to understand as possible.

By integrating the inference data with the interactive interface, the dashboard allows for:

- Dynamically adjusting probability thresholds to observe real-time changes in sensitivity and precision.
- Visually see the detection accuracy through colour-coded MRI slice overlays, categorising outputs into True Positives (green), False Positives (red), and False Negatives (yellow).
- Review a comprehensive dataset summary table that aggregates performance metrics across all 72 subjects, as well as having chart representations of the model performance.

2 Background

The task of cerebral microbleed detection spans two primary domains. The first is clinical neuroimaging, which covers the specific MRI modalities and cross-sectional planes critical for characterising such brain lesions. The second is artificial intelligence, specifically the application of deep learning convolutional frameworks such as nnU-Net, which enables highly accurate automated detection of these lesions. This section outlines the clinical and technical aspects necessary to help understand the methodology implemented in this study.

2.1 Magnetic Resonance Imaging for Microbleed Detection

Cerebral microbleeds (CMBs) are millimetre-sized phenomena that are purely invisible on standard/unweighted clinical scans such as computed tomography (CT) and many conventional MRI scans. To detect them, radiologists rely mainly on $T2^*$ -weighted sequences.[5, 15]

The $T2^*$ -weighted images are the critical sequence because they are highly sensitive to iron in old blood spots, and microbleeds will appear as rounded dark spots.[15, 5]

However, detecting CMBs is a challenging task due to the issue of look-alike phenomena. Anatomical structures such as blood vessels are visually identical to CMBs on $T2^*$ sequences, and if the model were to train solely on these, we would have a very high false positive rate, resulting in a low precision model. To mitigate these false positives, sequences such as T1 and T2 weights are used to help differentiate between actual microbleeds and look-alikes. The high risk of misclassifying such look-alikes further strengthens the need to have robust evaluation metrics for the model results, as well as not to over-rely on these metrics and be able to visually analyse the detections, as done so in the Interactive Dashboard Viewer.[5]

2.2 Automated Detection and the nnU-Net Framework

A traditional approach to CMB detection involves researchers doing manual feature engineering or spending a very long time during hyper-parameter tuning of convolutional neural networks. This process is highly sensitive to the variations in medical datasets, e.g.

voxel spacing, MRI resolutions and bleed counts.[7]

The nnU-Net was utilised for exactly this purpose. nnU-Net served as a highly reproducible and robust baseline due to it being a self-configuring deep learning-based segmentation model [6]. Rather than requiring the need for constructing a complex segmentation architecture from scratch, nnU-Net automatically adapts a standard U-Net architecture to the specific properties of the provided dataset.

For the VALDO Challenge dataset, the nnU-Net framework automates most of the preprocessing steps, including intensity normalisation, image resampling, and data augmentation [6]. For this project, a 2D full-resolution convolutional configuration was used due to training time constraints. Using nnU-Net as the baseline eliminated the extensive manual hyper-tuning required. This allowed the primary focus of this research to remain on evaluating the model’s outputs and on developing the Interactive Exploration Dashboard to interpret its clinical viability.

3 Dataset Description

3.1 Overview and Composition

The dataset utilised for this project originates from the Where is VALDO - Vascular Lesions Detection Challenge 2021, which was hosted at the 2021 International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). The central objective of this challenge was to promote the development of robust automated segmentation solutions for the identification of very small and sparse objects using weak and noisy labels. Ultimately, the challenge aimed to improve the early identification and diagnosis of cerebral small-vessel disease (CSVD) on brain MRI scans. Specifically, this project leverages the dataset provided for Task 2 of the challenge, which focuses exclusively on the detection of cerebral microbleeds.[3, 2, 15]

For every subject in the dataset, there are four files. Three of these files correspond to structural MRI volumes (T1, T2, and $T2^*$ -weighted sequences), while the fourth is the binary segmentation mask, where for each subject every microbleed was manually annotated by experts in the field. The segmentation mask shares the exact same dimensions as all MRI volumes, where every voxel is labelled as either 0 (denoting the background) or 1 (indicating a microbleed). All files within the dataset are of the NIFTI format (.nii.gz). As this is the standard file format for neuroimaging and medical artificial intelligence, it allowed for the integration of various established Python libraries to aid the project’s development.[2]

3.2 Licensing and Ethics

The VALDO dataset was released under the CC BY-NC-SA 4.0 license, which permits noncommercial research use. An academic dissertation at the University of Malta qualifies under these terms. Furthermore, because the dataset is already fully anonymised and publicly released, no ethics applications were required for this research.

3.3 Dataset Preprocessing

To prepare the dataset for training, the raw data had to be strictly formatted and restructured to meet the input requirements of the nnU-Net framework. Although nnU-Net is a self-configuring model, it requires a highly specific directory structure and naming con-

vention. Specifically, each dataset must reside within an `nnUNet_raw/DatasetXXX_Name` directory. Within this directory, training images are placed in an `imagesTr` folder and must follow the naming convention `{CASE_IDENTIFIER}_{XXXX}.{FILE_ENDING}`, where `XXXX` represents a four-digit modality suffix. Corresponding training labels are placed in a `labelsTr` folder and are named simply `{CASE_IDENTIFIER}.{FILE_ENDING}`. Optional test images can be stored in an `imagesTs` folder. Finally, a `dataset.json` metadata file must be included at the root of the dataset folder.[6]

A Python script was created to establish a data preparation pipeline to read the original dataset directory, rename all the files and organise them into the nnU-Net structure. Additionally, during this process, alignment was performed to align the labels with the $T2^*$ -weighted volume to ensure spatial accuracy before feeding the data to the model. Under this required naming convention, each file uses a four-digit modality suffix to differentiate the sequences, `_0000` for T1, `_0001` for T2, and `_0002` for $T2^*$. Finally, the `dataset.json` file maps these modality suffixes to human-readable names, defines the segmentation classes, and specifies the expected file extensions.

3.4 Label Verification

Before initiating the training process, it was necessary to verify that all labels were correct and consistent. Training on misaligned or corrupted labels will result in a model that produces misleading and meaningless results. To prevent this, a verification pipeline was implemented to perform various checks:

- **Label Value Integrity:** Ensuring that every voxel within every label file was strictly assigned a value of either 0 or 1.
- **Spatial Alignment:** Verifying that the label volumes have the same shape dimensions and spatial mapping (affine matrix) as their corresponding image volumes.
- **Connected-Component Statistics:** Utilising SciPy to calculate the number of discrete microbleed objects per subject, identifying how many microbleeds exist per subject, and evaluating their individual voxel sizes to establish initial dataset statistics.

3.5 Dataset Characteristics and Statistics

Following the label verification process, a statistical analysis of the ground-truth segmentations was conducted to help understand the dataset’s distribution. Documenting such characteristics was critical to the project as the variation of frequencies and sizes of microbleeds could heavily influence both the model’s training dynamics, as well as the final evaluation metrics.

Figure 1 illustrates the distribution of microbleed counts across all 72 subjects. The dataset shows significant variance; while most subjects possess only a few microbleeds (e.g. 1-5), a subset of severe cases contains 20 or more. The variation in the lesion counts shows that the subjects are varied enough for the model to be able to generalise well during the training process.

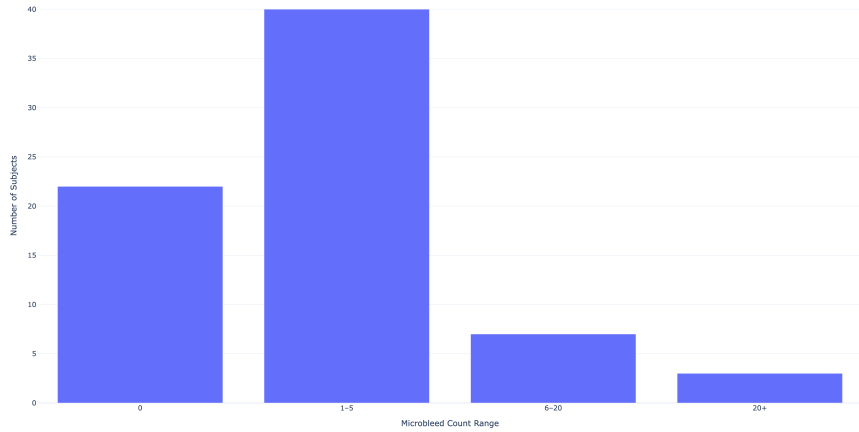


Figure 1: Distribution of the total microbleed counts per subject across the VALDO training dataset.

Furthermore, the physical volumes of the individual microbleeds were calculated to document their size distribution. This was achieved by taking a voxel count of each verified connected component and then converting it into cubic millimetres (mm^3) using voxel spacing extracted from the NIfTI file headers.

From Figure 2, it is clear that the lesions are exceptionally small. This highlights the main challenge that the nnU-Net faces when attempting to distinguish the millimetre-sized lesions from the vast background of healthy brain tissue.

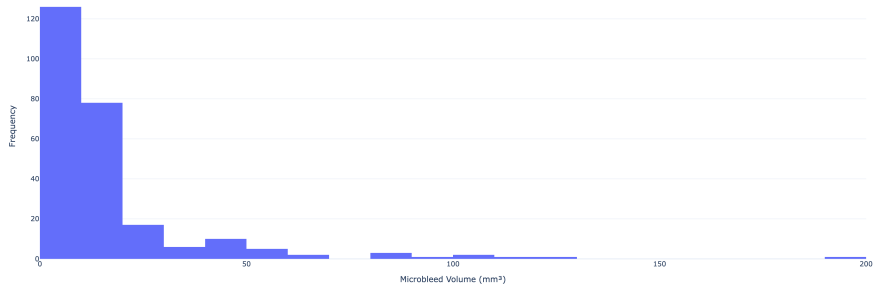


Figure 2: Histogram of individual microbleed sizes (in mm^3)

4 Shared Baseline Pipeline

4.1 Methodology

The shared baseline pipeline is concerned with the utilisation of the nnU-Net framework to perform automated detection of cerebral microbleeds. Alongside this is an evaluation pipeline to extract custom evaluation metrics from the model inference stage.

Figure 3 illustrates the architecture of the completed pipeline, divided into three stages. The first stage is the preprocessing pipeline, which was described in detail in Section 3.5. The remaining stages correspond to the nnU-Net pipeline, encompassing the framework’s internal preprocessing, model training, and inference steps. The final stage is the Interactive Exploration Dashboard.

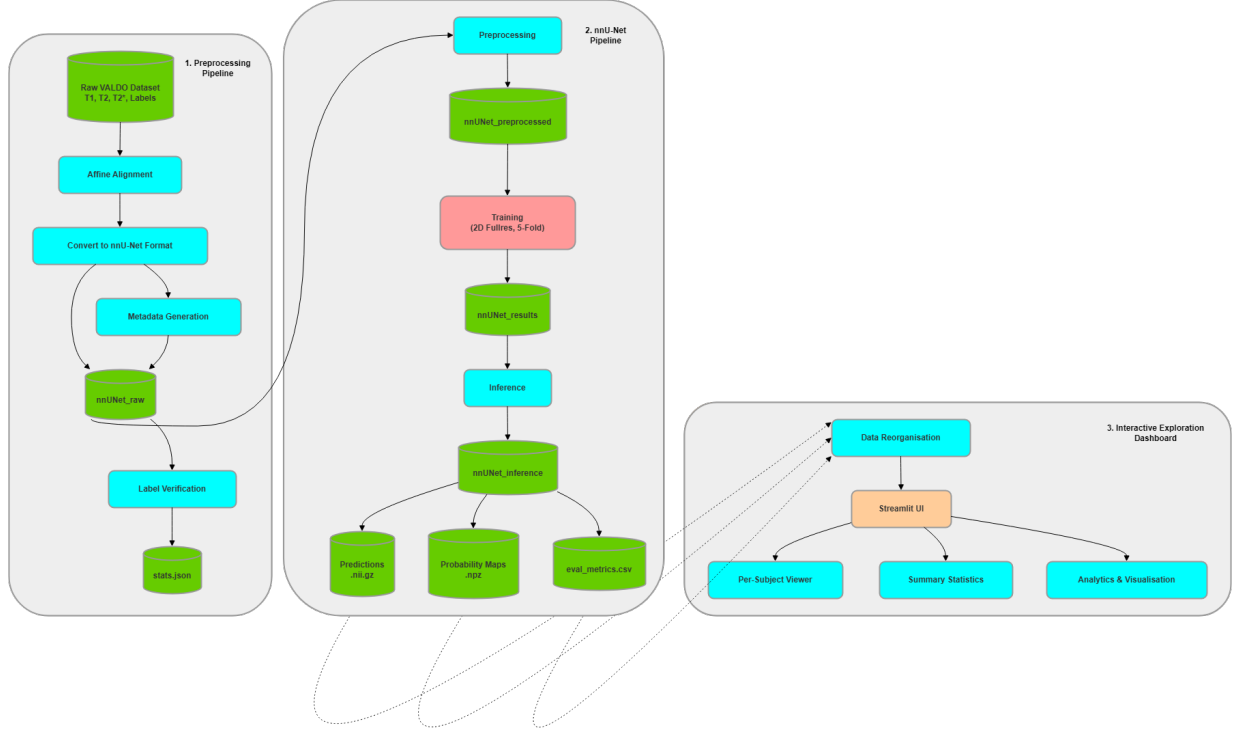


Figure 3: System architecture detailing the preprocessing pipeline, the nnU-Net framework processing stages, and the Interactive Exploration Dashboard.

4.1.1 Baseline Model Training

To establish a robust and reproducible baseline for microbleed detection, the nnU-Net deep learning framework was implemented [6]. Since training was performed on three weighted

volumes (**T1**, **T2**, $T2^*$) alongside segmentation masks of MRI slices, the task of cerebral microbleed detection is computationally expensive. For this reason, the 3D full-resolution configuration of nnU-Net was avoided, as it would significantly increase training time. Instead, the **2D full-resolution** configuration was selected as a practical alternative, expected to achieve strong results with minimal performance loss despite the lack of inter-slice contextual information.

To ensure the generalizability of the model and rigorously evaluate its performance on unseen data, a **five-fold cross-validation** strategy was applied to the VALDO training dataset. To ensure a reliable shared baseline was kept, the **standard nnU-Net data splits** were performed.

4.1.2 Inference and Standardised Output Generation

Once the nnU-Net model was trained, an inference pipeline was executed to generate predictions on the dataset. To ensure compatibility with the interactive dashboard, the outputs were standardised into three specific formats:

1. **Prediction Masks:** NIfTI files (`.nii.gz`), one per subject, with the same case identifier as the original dataset. Binary segmentation masks (0/1) at the default nnU-Net threshold.
2. **Probability Maps:** NumPy array files (`.npz`) containing the raw softmax confidence probabilities for the microbleed class, enabling dynamic thresholding for the interactive dashboard.
3. **Evaluation Metrics:** A structured CSV file (`evaluation_results.csv`) containing the quantified performance metrics per subject.

4.1.3 Evaluation Logic and Connected-Component Analysis

Because microbleeds are sparse phenomena, relying solely on standard voxel-wise metrics (such as the Dice coefficient) can be misleading, as even a single misplaced voxel can significantly skew the score for very small objects. To address this, an evaluation script was developed to perform lesion-level connected-component analysis.

The lesion-level evaluation was implemented as a per-subject analysis, where each ground-truth label is matched with its corresponding predicted mask. Both volumes were

loaded and binarized, followed by 3D connected-component analysis using a $3 \times 3 \times 3$ **connectivity structure** to isolate individual microbleeds. A **many-to-one overlap strategy** was adopted such that ground-truth components overlapping with any predicted voxels were counted as True Positives, while predicted components with no overlap were classified as False Positives, and missed ground-truth components as False Negatives. Lesion-level sensitivity and precision were computed from these counts, with the F1-score defined as the harmonic mean of these values. In parallel, a voxel-wise Dice coefficient was calculated to quantify spatial overlap. This process was applied across all subjects and was exported as a per-subject CSV file for later analysis and visualisation.

The following list of metrics is what was recorded during the evaluation process:

- **Lesion-Level Sensitivity (Recall):** This measures the proportion of actual microbleeds successfully identified by the model. A microbleed is considered a True Positive (TP) if the predicted mask overlaps the ground-truth microbleed by at least one voxel. Missed microbleeds are False Negatives (FN).

$$Sensitivity = \frac{TP}{TP + FN} \quad (1)$$

- **False Positives per Subject:** Number of incorrectly detected microbleeds by the model. A microbleed is considered to be a false positive if its corresponding connected component in the predicted mask has no spatial overlap with any ground-truth microbleed.
- **Lesion-Level F1-Score:** The F1-score provides us with a harmonic mean between the lesion-level precision and sensitivity, serving as a very strong metric to get the overall performance of the model within one value.

$$F1 = 2 \times \frac{Precision \times Sensitivity}{Precision + Sensitivity} \quad (2)$$

- **Voxel-Level Dice Coefficient:** While the above metrics evaluate performance at the object level, the Dice similarity coefficient evaluates the exact spatial overlap at the voxel level. Let X be the set of ground-truth voxels and Y be the set of predicted voxels:

$$Dice = \frac{2|X \cap Y|}{|X| + |Y|} \quad (3)$$

- **Number of Ground Truth Microbleeds:** Total number of ground truth microbleeds for a given subject.
- **Number of Predicted Microbleeds:** Total number of predicted microbleeds by the model for a given subject.

4.2 Results

4.2.1 Baseline Results

The evaluation of the baseline nnU-Net model using the VALDO dataset shows very promising results, especially in terms of precision. As illustrated in the scatter plot Figure 4, there is a significant positive correlation between the ground-truth microbleed counts and the predicted counts. The model displays exceptional accuracy for subjects with low microbleed counts. However, for severe cases, such as for subject VALDO_110, which contained 73 ground-truth microbleeds, the model exhibits a slight tendency to have a higher false negative count, resulting in a marginally lower sensitivity for extreme cases, but still manages to capture the majority of the bleeds.

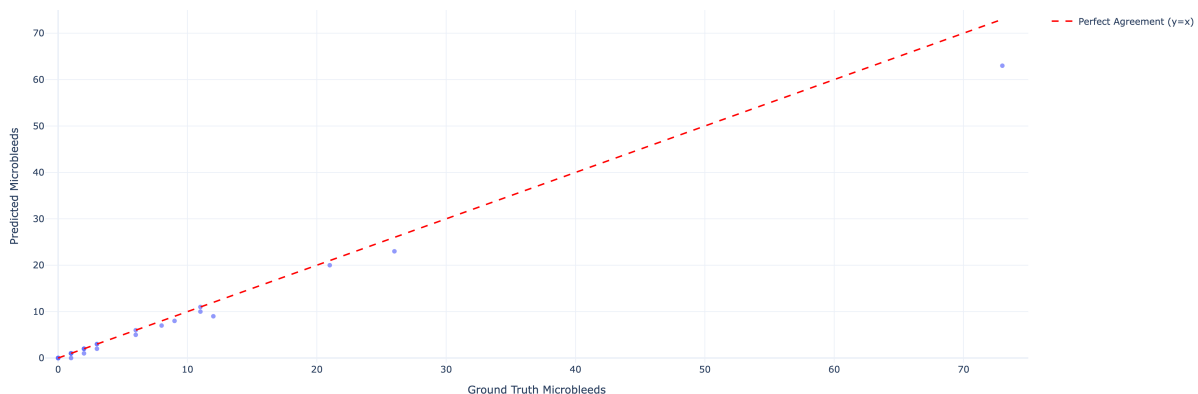


Figure 4: Scatter plot of Ground Truth vs. Predicted Microbleed Counts per Subject.

Taking a closer look at the performance of the model, as presented in the boxplot Figure 5, we can get a stronger idea of where our model lies. The median values for Lesion-Level Sensitivity, F1-Score, and the Voxel-Level Dice Coefficient are all concentrated near 1.0. One of the most interesting observations that was made has to do with the False Positive rate, which was negligible across all evaluated subjects. The most volatile

metric was the Dice coefficient, which is mathematically expected as a single misclassified voxel significantly penalises the spatial overlap metric for tiny structures like cerebral microbleeds, even if the lesion was successfully detected at the object level.



Figure 5: Boxplot distribution of Lesion-Level Sensitivity, F1-Score, and Voxel-Level Dice Coefficient across all subjects.

4.2.2 Aggregate Performance Statistics

To quantify the model’s performance across the entire dataset of 72 subjects, we can view the statistical summary table found in Table 1. The most remarkable finding from Table 1 is that the model produced **zero false positives** across all subjects, resulting in perfect precision.

Both **Sensitivity** and the **F1-Score** achieved a **median of 1.0**, indicating a perfect lesion-level detection for the majority of the dataset. The **mean Sensitivity (0.960)**, **mean F1-Score (0.971)** and **mean Dice coefficient (0.895)** are just slightly lower than their respective medians, indicating that the place where the model suffered most was at the rare edge cases where the model failed to detect highly clustered or exceptionally small microbleeds.

Metric	Mean	Std. Dev.	Min	Median	Max
Sensitivity	0.960	0.141	0.000	1.000	1.000
F1-Score	0.971	0.127	0.000	1.000	1.000
Dice Coefficient	0.895	0.160	0.000	0.977	1.000
Ground-Truth Microbleeds	3.28	9.46	0	1	73
Predicted Microbleeds	2.94	8.24	0	1	63

Table 1: Summary statistics of model performance across all 72 subjects.

4.3 Discussion

The evaluation of the baseline nnU-Net model showed that the 2D full-resolution configuration is highly capable of accurately detecting cerebral microbleeds. Most notably, the model achieved a perfect precision as it generated no false positives across all inference subjects. As both sensitivity and F1-score reached a median of 1.0, the baseline model achieves perfect lesion-level detection for the standard clinical cases.

The results also showed that there is a fundamental limitation of relying on the traditional spatial overlap metrics for such small-object detection. The voxel-level dice coefficient experiences a higher volatility and a slightly lower mean of 0.895 compared to object-level metrics. This is expected, as a single misclassified voxel heavily penalises the Dice score for millimetre-sized structures, even when the predicted lesion is at the correct spatial location.

While the model generalises exceptionally well for subjects with normal lesion counts, it struggles more with severe outliers. For instance, in extreme edge cases like subject VALDO_110, which contained 73 ground-truth microbleeds, the model exhibited a higher rate of false negatives but still a promising result by predicting 63 correct microbleeds and only having 10 false negatives. This indicates that while the 2D architecture is efficient and highly precise, the detection of **tightly clustered** or **exceptionally small** microbleeds in severe cases is still a challenging task for achieving a near-perfect prediction.

5 Individual Research Thread: Interactive Exploration Dashboard

5.1 Methodology

To bridge the gap between aggregate statistical metrics and qualitative spatial analysis, an Interactive Exploration Dashboard was developed. The core aim of this research thread is to provide researchers and clinicians with an interface to inspect individual test subjects slice-by-slice. Furthermore, it allows users to visually evaluate the model’s predictions and observe how dynamic adjustments to the probability threshold directly impact the network’s predictive capabilities.

The application was built using a modular architecture in Python. The data ingestion pipeline, managed via a **reorganisation script**, automatically extracts the NIfTI MRI volumes, binary masks, raw SoftMax probability maps and the evaluation summary from the nnU-Net inference outputs and reconstructs them in the accepted format for the interface. Figure 3

The backend processing relies heavily on **NiBabel** for managing neuroimaging data. A custom NIfTI loader extracts the 3D voxel arrays alongside their affine matrices to ensure accurate spatial orientation. To evaluate performance at the lesion level, a connected-component matching algorithm was constructed and follows a similar implementation to that in Section 4.1.3 to ensure accurate calculations throughout the entire project. The algorithm extracts distinct components from the prediction masks and evaluates their spatial intersection with ground-truth components to classify each prediction dynamically.

The frontend user interface was developed using the **Streamlit** framework, chosen for its implementation simplicity and modern features. The dashboard is split into three views: the per-subject slice viewer, a dataset summary section, and graph analysis.

5.1.1 Per-Subject Slice Viewer

The core component of the dashboard is a multi-planar slice navigation interface, allowing users to traverse the 3D MRI volume across the **axial, sagittal, and coronal** planes. To aid qualitative analysis, segmentation masks are rendered as colour-coded overlays directly

on top of the rendered slice, with **true positives shown in green, false positives in red, and false negatives in yellow.**

A key feature implemented within this viewer is the **dynamic probability thresholding**. Using a threshold slider ranging from 0.0 to 1.0, users can **dynamically adjust the threshold** applied to the **raw SoftMax probability maps**. Updating this threshold triggers a real-time recalculation of the connected components, instantly refreshing the overlay visualisations. Users are also able to toggle the segmentation masks on and off to directly compare the model’s predicted segmentations against the underlying microbleeds.

To further connect these spatial visualisations with quantitative evaluation, a dedicated metrics section is displayed alongside the slice viewer. This section holds subject-level detection statistics, including the counts of **true positives, false positives, and false negatives**, as well as the total **number of ground-truth and predicted microbleeds**. Additionally, the performance metrics (**sensitivity, precision, and F1-score**) are also included. An important note is that each of these metrics are immediately updated the second you adjust the probability threshold to give a dynamic experience of the interface.

5.1.2 Dataset Summary and Analytics

The remaining two sections focus on the analytical components of the interface.

The macro-level analysis is presented within the **Dataset Summary** section, where an interactive dataframe was added to display the evaluation metrics for all subjects, including **F1-score, sensitivity, Dice coefficient, false positives, ground-truth counts, and predicted counts**. Alongside this table, an evaluation overview was included to provide aggregated statistics for the key metrics, namely F1-score, sensitivity, Dice coefficient, and false positives. Additionally, a key feature was implemented where users can also select a subject directly from the interactive table, and it will automatically update the per-subject viewer to display the corresponding scans and predictions.

The visual analysis component is presented in the **Graph Analysis** section, where two dynamic charts were implemented using Streamlit’s charting libraries: the microbleed count per subject and the sorted F1-score per subject.

The first chart provides a frequency distribution of the ground-truth microbleeds across all inference subjects. This visualisation was mainly added to highlight class imbalances within the dataset.

The second chart plots the F1-score of each subject in ascending order. By sorting the data from worst to best performance, this chart allows the user to quickly interpret the model’s overall consistency. The main purpose of this chart was to support the rapid identification of poorly performing subjects so that they could then be examined in greater depth in the **Dataset Summary** and **Per-Subject Viewer** sections.

5.2 Results

The integration of the Interactive Exploration Dashboard with the validation allowed for the transformation of the evaluation process from a static statistical summary into a dynamic and clinically interpretable diagnostic review. While the aggregate metrics (shown in Table 1) provide a meaningful high-level overview of model performance, they often fail to fully capture the sensitivity and complexity associated with CMB detection. By leveraging the dashboard’s architecture, a more human-interpretable analysis of the model’s predictive ability became possible, revealing subtle details in probability calibration and small-object localisation that were previously overlooked. The following section, therefore, discusses both qualitative and quantitative insights obtained through a more detailed inspection of the model’s predictive performance.

5.2.1 Qualitative Analysis: Per-Subject Slice Viewer Tab

The per-subject viewer shown in Figure 6 enabled a direct spatial verification of the model’s detections against the ground-truth annotations. By implementing a real-time connected-component matching algorithm, the dashboard provided immediate visual feedback through colour-coded prediction overlays.

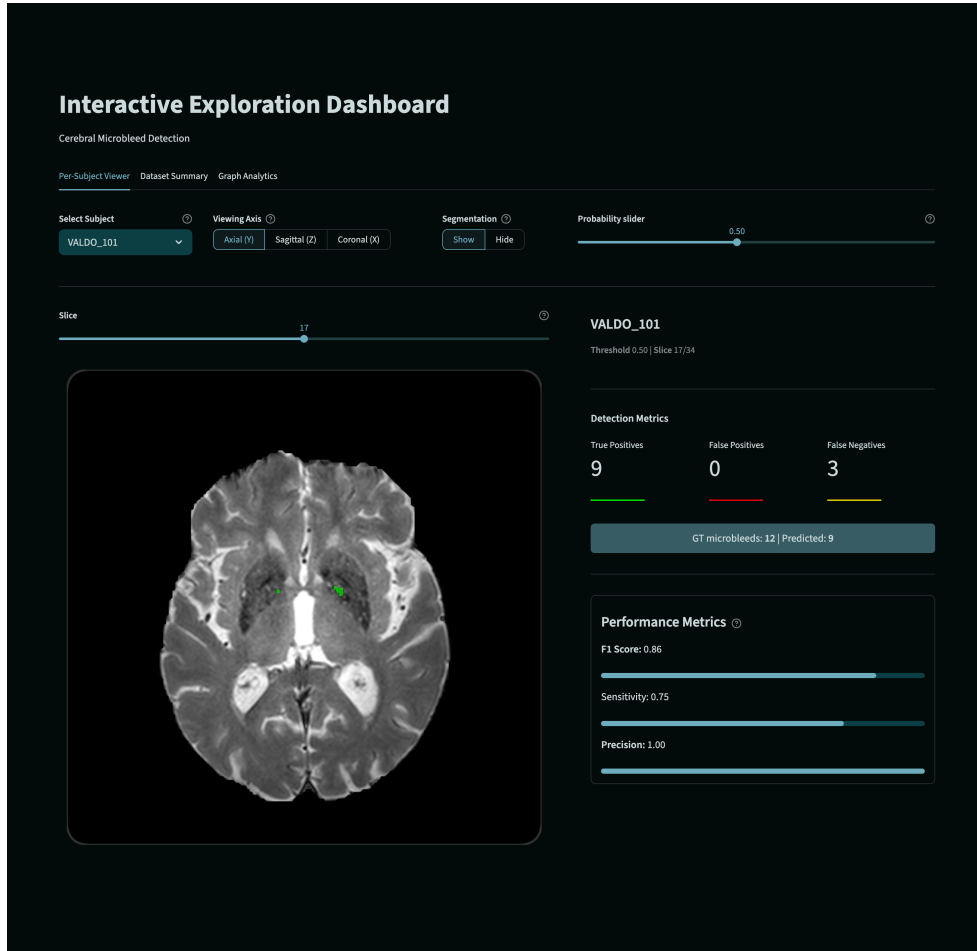


Figure 6: The Per-Subject Slice Viewer interface. In this instance, we have subject VALDO_101 at slice 17 with two correctly predicted lesions highlighted in green.

A key finding facilitated by this tool was the observation of **sub-threshold localization**. Through the use of the dynamic probability slider, it was discovered that the baseline model’s False Negatives could be significantly reduced by simply setting the Soft-Max probability threshold to be slightly less strict than the default of 0.5. This value, through empirical testing, was often in the range of $P(x) \in [0.25, 0.40]$ where, at this value from the subject viewer, the overlays of the False Negatives (yellow) are instantly converted to True Positives (green). This result suggests that what the model considers as failures are often a matter of probability calibration rather than the model failing to capture the necessary features, indicating that a slightly less strict threshold could be beneficial to the model performance without having the risk of higher false positive counts.

5.2.2 Quantitative Analysis: Dataset Summary Tab

The dashboard’s interactive performance table (Figure 7) was integrated with the main scope for triaging the validation subjects.

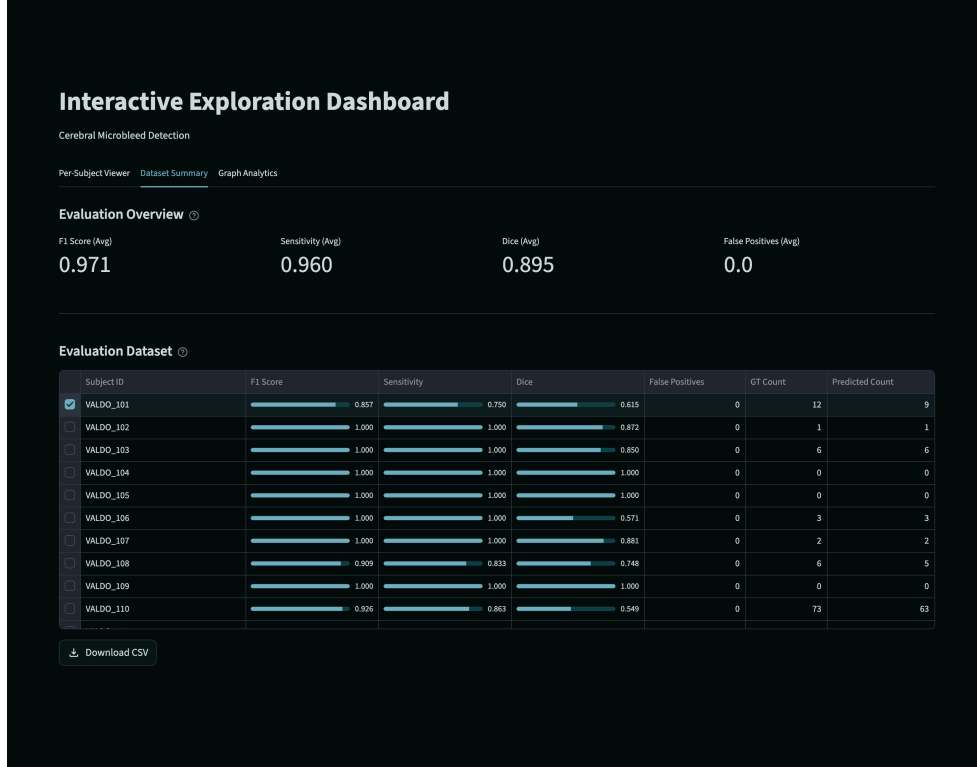


Figure 7: The Dataset Summary tab. Includes all key metrics for each subject and allows for sorting by any desired metric

The facility of sorting subjects by key metrics, such as F1-score and sensitivity, allowed for quick identification of poor-performing subjects. Specifically, the Dataset Summary Tab indicated that the drop in performance was not purely random but was concentrated around edge case subjects where either they had extremely small, single-voxel lesions or extreme counts of microbleeds. By identifying such subjects in the table, the researcher could easily select the problematic subjects and analyse them in the Per-Subject Slice Viewer.

5.2.3 Quantitative Analysis: Graph Analysis Tab

The graphical analysis section (Figure 8) provided a macro-level perspective on the model’s consistency across the varying distributions of the VALDO dataset.

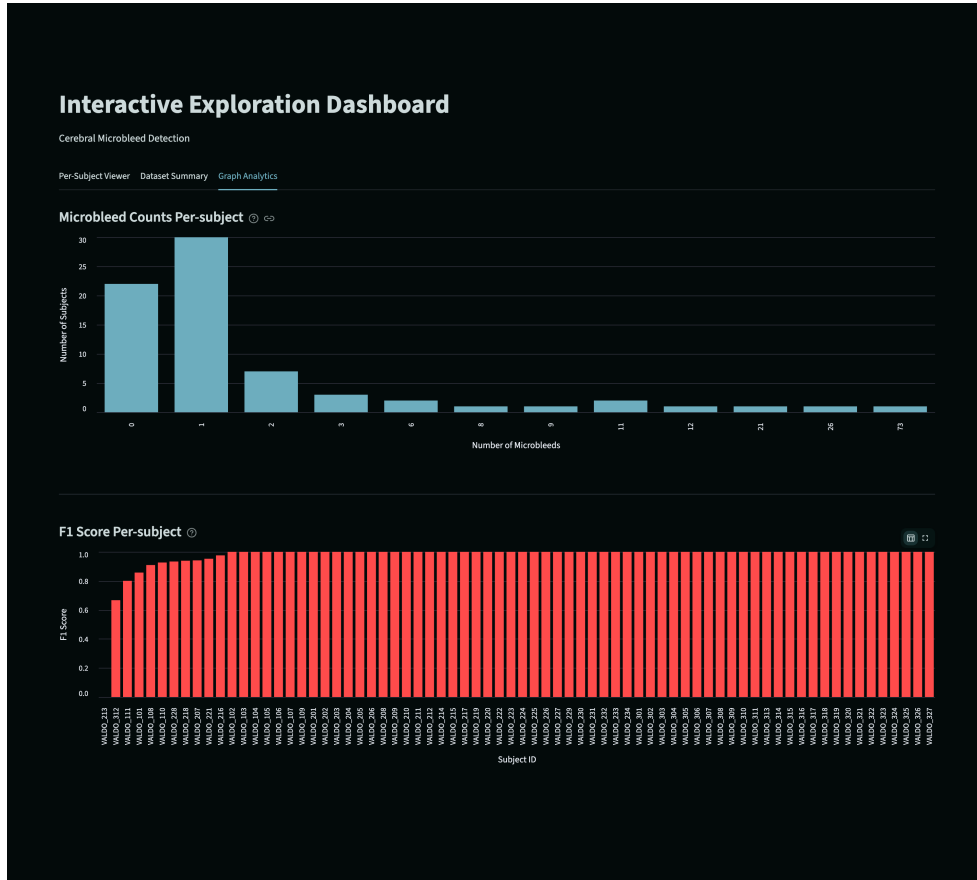


Figure 8: The Graph Analysis Tab. Illustrates the frequency distribution of ground-truth microbleeds (top graph) and an ascending ordered ranking of per-subject F1-Scores (bottom graph).

The distribution of microbleed counts (top chart) verifies that the majority of the subjects are skewed towards the lower end of the microbleed spectrum at **around zero to two lesions**. The corresponding F1-score rankings (bottom chart) show that the model is exceptionally strong on the standard clinical cases, with the **majority of the inference subjects achieving a perfect F1-score**. This chart also further clarifies that the model struggles most not with the general cases but with the outliers in the dataset, which as a result occupy the lower end of the F1 Score rankings.

5.3 Discussion

The dashboard results demonstrate that for very small object detection tasks, such as our CMB detection, simple voxel-wise metrics such as Dice are insufficient for a complete evaluation. This tool provides the necessary evidence that even when a model’s Dice score is

heavily penalised by the small size of the microbleed, its predicted location in the MRI volume is often highly accurate. This approach demonstrates that interactive visual analytics are mandatory for standard deep learning evaluation to provide clear transparency of the model results, thereby strengthening trust in automated AI diagnostic systems.

6 Conclusions

This research successfully implemented a robust automated pipeline for the detection of cerebral microbleeds (CMBs), overcoming many limitations faced by manual CMB detection. By utilising a 2D full-resolution self-configuring nnU-Net model, the baseline model achieved exceptional performance, producing zero false positives across all inference subjects alongside a perfect median lesion-level sensitivity and F1-score. The model proved to be very effective for traditional clinical cases, with only a slight decrease in sensitivity in severe edge cases where the subjects would have an exceptionally high microbleed count bound tightly next to each other.

Furthermore, this study highlighted the limitations of traditional spatial overlap metrics, such as the voxel-level Dice coefficient, which often fails to capture the true predictive ability of the model, especially for small-object detection, as is CMBs detection. To address this issue, an Interactive Exploration Dashboard was developed to enable visual analysis of model predictions for each subject. Through interactive probability threshold adjustment, several false negative predictions were converted into true positives, indicating that the model actually manages to capture the necessary features of the microbleeds but occasionally requires some minor calibrations to the probabilities.

In conclusion, the nnU-Net framework, along with the Interactive Exploration Dashboard, provides a high-performing and transparent baseline for automated clinical diagnostics. Additionally, an important observation is that the methodology developed within this research has the potential to extend beyond the brain MRI applications. Since the pipeline is specifically optimised for small-object detection, both the model architecture as well as the Interactive Exploration Dashboard can be adapted to other medical imaging domains, in fact, this is one of the main reasons why nnU-Net is such a strong baseline model [6]. Some applications where this would be relevant are for detecting phenomena like **Pulmonary Nodule** (areas of abnormal growth on your lungs), **Bone Microfractures**, and **Osteolytic Bone Metastases** (distant tumour deposits of a primary tumor within bone) [4, 13, 12]. Future work ideally should explore these broader applications to the model, as well as investigating 3D model configurations to analyse the gap in model performance due to it having inter-slice context, as well as further improvements with detections amongst outlier subjects with highly clustered cases.

7 Use of Generative AI

7.1 Tools Used

- Google Gemini (Mainly used for documentation and clarification of logic)
- OpenAI ChatGPT (Mainly used for debugging and implementation ideas)

7.2 Where Generative AI Was Used

Generative AI was used in the following areas:

- **Brainstorming:** Generative AI was used to help formulate and refine initial ideas.
- **Code Implementation and Debugging:** Generative AI was used to assist in converting implementation ideas into code, of which I was unsure how to implement correctly. It was also used to support the development of the Streamlit dashboard and to guide debugging.
- **Concept Explanation:** Generative AI was used to explain difficult and unfamiliar concepts within the field of medical imaging.
- **Documentation:** Generative AI was used to help maintain a formal and academic tone throughout the documentation.

7.3 How it is Used

AI was strictly used and restricted as a tool, and it was ensured that it did not replace the idea stage.

For code generation and debugging, code was not simply copied and pasted within the project; rather, prompts to aid with code segments were utilised, and the results were reviewed. If the logic made sense, the idea was used. During the coding stage, generative AI was especially useful during the implementation of the Streamlit dashboard, where it helped generate repetitive code that was common throughout the UI. It was also useful in the section where I had to choose how to identify the true and false positives and true and false negatives, where at the end, the final decision was to implement a many-to-one overlap strategy for mapping our predictions to the ground truth mapping our predictions

to the ground truths.

For the documentation stage, the AI helped by suggesting ideas for where, in my written paragraphs, I could make them more formal and more academic, but throughout I constantly ensured that it would not suggest a completely different tone. It was also used to clarify concepts for this project which I was unfamiliar with or not confident in. The responses were cross-checked with other online sources to ensure that I would not be prone to hallucination mistakes.

7.4 Verification

The AI outputs were verified constantly to ensure that the final output is relevant, accurate and that the final submission would be my own. Any AI-generated code suggestions were not added blindly to the project but were reviewed carefully. The way code generation was utilised was as suggestions for additional implementation to the project, and then adopt code ideas from that to ensure that the code implementation would be my own.

For documentation ideas or concept clarifications, the documentation was compared to various sources; this way, I ensured that the ideas that I took would not be off-topic and would not be prone to consuming information from hallucinations.

Overall, Generative AI was used as a support tool only, while the final decision and implementation were solely done by me.

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